

Global Aviation Disruption Assessment 2026

Preliminary Data Assessment

Initial assessment of the supplied data, its analytical coverage, and the decisions required before cleaning and reporting.

Scenario note: This is a simulated ICAO client assignment, not an official ICAO commission or endorsement.

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1 Assessment conclusion

The seven files support descriptive operational, geographic, financial, and event-context analysis. They do not form one common flight population. The main preparation decisions are to preserve all records, retain the distinct financial populations, standardize three airline aliases, preserve full closure timestamps, and retain conflict locations without inventing countries.

The following checks examine the supplied raw files and compare them with the implemented clean outputs. They establish the preparation decisions; they do not represent a historical client approval or a claim of complete real-world coverage.

2 Supplied datasets and coverage

Source	Rows and grain	Date coverage
Modeled airline summary	70; One row per airline	Undated
Airline daily estimate	35; One row per airline	Undated
Airport disruptions	227; One disruption record	2026-02-27 to 2026-06-07
Airspace closures	84; One closure record	2026-02-28 to 2026-06-07
Conflict context	82; One contextual event	2025-12-15 to 2026-06-08
Detailed cancellations	2,200; One cancellation record	2026-02-27 to 2026-06-07
Detailed reroutes	1,500; One reroute record	2026-02-28 to 2026-06-07

Airspace closure ends extend to 17 June 2026. Conflict context begins on 15 December 2025, earlier than operational records. Operational comparisons therefore need a narrower common period. The financial files are undated, even where a field is described as a daily estimate.

2.1 Entity coverage

Source	Distinct source entities
Modeled airline summary	airline: 70; country: 48; region: 7
Airline daily estimate	airline: 35; country: 29
Airport disruptions	iata_code: 55; country: 44; region: 8
Airspace closures	country: 25; region: 6
Conflict context	country: 14
Detailed cancellations	airline: 70; origin_country: 43; destination_country: 43
Detailed reroutes	airline: 70; origin_country: 43; destination_country: 43

3 Initial data quality checks

Source	Raw missing cells	Exact duplicate rows	Rows retained
Modeled airline summary	0	0	70 / 70
Airline daily estimate	0	0	35 / 35
Airport disruptions	0	0	227 / 227
Airspace closures	0	0	84 / 84
Conflict context	0	0	82 / 82
Detailed cancellations	0	0	2,200 / 2,200
Detailed reroutes	0	0	1,500 / 1,500

No raw field is empty and no exact duplicate row is found in the seven files. These checks do not prove that the snapshot is complete or that every flight is unique. Repeated flight numbers are examined separately because the same number can occur on different dates or records.

Detailed source	Rows with repeated flight numbers	Repeated date airline flight combinations
Detailed cancellations	18	0
Detailed reroutes	8	0

All examined count, cost, distance, and duration fields are nonnegative. Revenue-loss values lie between 3.6% and 53.4%. The supplied reroute distance arithmetic and closure-duration arithmetic pass the implemented tolerances. Airport IATA codes are checked for consistent airport and country mapping. Valid Unicode text is retained.

4 Population reconciliation

Comparison	Result	Decision
Cancellation summary versus detail	6,030 summary counts; 2,200 detail records	Keep both; do not treat the summary as a detail aggregation.
Reroute summary versus detail	4,539 summary counts; 1,500 detail records	Keep both; do not force counts to match.
Financial airline coverage	70 summary; 35 estimate; 32 shared; 73 union	Retain estimate-only and summary-only airlines.
Airport versus airspace exposure	35,844 airport flights; 35,502 airspace flights	Report separately; overlap cannot be deduplicated.

Comparison	Result	Decision
Financial date coverage	No dates in either financial file	No date relationship or period extrapolation.

All seven files came from the same supplied dataset snapshot. The files do not document a common scope that would explain the different financial and detailed totals. Different populations are therefore the reporting rule, not a proven explanation of how the source was generated. Neither table is declared wrong solely because its total differs.

Austrian Airlines, El Al, and Royal Jordanian are present only in the estimate file after name alignment. Thirty-eight summary airlines are absent from the estimate file. A shared airline name establishes entity overlap, not an identical event or financial population.

5 Required preparation decisions

Source	Preparation decision
All files	Trim surrounding text whitespace, check schema and required fields, preserve every row, and write UTF-8 outputs.
Airline summary	Retain source financial values and airline-type categories.
Airline estimate	Align PIA, Saudia, and Swiss International to their documented full airline names.
Airport disruptions	Parse dates, validate nonnegative measures, and check IATA mapping.
Airspace closures	Retain complete timestamps and validate end order and duration; do not invent timezone information.
Conflict events	Retain source_location and derive location_name and country only from the source delimiter.
Cancellation detail	Parse dates, validate passenger counts, and report repeated flight numbers without deleting records.
Reroute detail	Parse dates, validate numeric ranges and the distance formula, and preserve repeated flight numbers.

6 Conflict location assessment

The raw event file contains 82 nonempty locations. Forty-two cannot yield a country through the implemented location split. The clean file retains the supplied location text after trimming and leaves country null for those records. These are unresolved or non-country-specific records, not 42 confirmed missing-country errors.

Country-specific event analysis includes only assigned countries. Regional and global context remains available in the complete event timeline. No country is inferred from an event description.

7 Source authority for metrics

Business metric	Authoritative supplied population
Cancellation events and affected passengers	Detailed cancellations
Reroutes, distance, fuel cost, and delay	Detailed reroutes
Modeled total loss and revenue-loss percentage	Airline summary
Estimated daily loss, passengers, fuel, cancellations, and reroutes	Airline daily estimate
Airport exposure	Airport disruption records
Airspace exposure	Airspace closure records
Conflict and military context	Conflict event records

8 Business question coverage

Questions	Supported analysis	Boundary
Q01 to Q08	Detailed counts, passengers, reasons, dates, distance, fuel, and delay	No scheduled-flight denominator or reroute-passenger field.
Q09 to Q11	Directional corridor counts and burdens	Concentration is not a route risk rate.
Q12 to Q18	Airport, country, regional, duration, and closure comparisons	Airport and airspace exposures remain separate.
Q19 to Q23	Financial rankings and population differences	Undated modeled values are separate scenarios.
Q24	Daily military-event and other-date comparison	Use shared operational coverage; timing does not establish causation.

9 Analytical decision

Proceed with source-specific descriptive analysis. Use the operational intersection from 28 February to 7 June 2026 for event-date comparison and keep the wider calendar for context. Define every measure before presenting it. The completed analysis and recommendations belong in the Executive Case Study Report; implementation detail belongs in the Technical Documentation.